Dell PowerStore: Deploying Proxmox Virtual Environment

White Paper

Abstract

This white paper describes the deployment and configuration of Proxmox Virtual Environment on Dell PowerStore to create, manage, and monitor the virtual machines using shared storage. The paper also describes various virtualization tasks on virtual machines in Proxmox Virtual Environment. It also describes best practices for deploying the PostgreSQL database on Dell PowerStore in a Proxmox Virtual Environment.

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1

Executive summary

Proxmox Virtual Environment (VE) is an open-source platform that combines virtualization and containerization, providing a complete solution for managing workloads efficiently. It supports both Kernel-based Virtual Machine (KVM) for virtual machines and Linux Containers (LXC) for containers, enabling enterprises to run various applications on a single platform. Proxmox VE features a convenient web interface, high availability clustering, backup, and restore options, making it a flexible choice for both small and large environments. Its flexibility and powerful tools make it ideal for optimizing resources and streamlining operations.

The Dell PowerStore platform uses a container-based microservices architecture, advanced storage technologies, and integrated machine learning to unlock the power of data. It is a versatile platform with a performance-centric design that delivers multidimensional scale, always-on data reduction, and support for next-generation media. PowerStore brings the simplicity of public cloud to on-premises infrastructure, streamlining operations with an integrated machine-learning engine and seamless automation. It also offers predictive analytics to monitor, analyze, and troubleshoot the environment. PowerStore provides organizations with a data-centric, intelligent, and adaptable infrastructure that supports both traditional and modern workloads.

This white paper describes:

- The deployment and configuration of Proxmox Virtual Environment to create, manage, and monitor virtual machines.
- The configuration of iSCSI, NVMe/TCP, and NFS in Proxmox VE to connect and use the PowerStore system.
- Various virtualization tasks such as templates, cloning, live migration, storage migration, and snapshots of VMs running on Proxmox VE.
- The best practices for deploying the PostgreSQL database using the PowerStore system in a Proxmox VE environment.

Introduction

Audience

This white paper is primarily intended for sales engineers, field consultants, IT administrators, and customers. It is also intended for anyone else who is interested in configuring and deploying a PostgreSQL database in Proxmox VE on the PowerStore system.

Readers are expected to have an understanding and working knowledge of Proxmox VE, Virtual Machines, PostgreSQL databases and PowerStore.

Terminology

The following table provides definitions for some of the terms that are used in this document.

Table 1. Terminology

Term	Definition
DNS	Domain Name System
FQDN	Fully qualified domain name
iDRAC	Integrated Dell Remote Access Controller
KVM	Kernel-based Virtual Machine
LCM	Lifecycle Management
LXC	Linux Containers
LVM	Logical Volume Manager
NFS	Network File System
NIC	Network Interface Card
NVMe	Non-Volatile Memory express
NVMe-oF	NVMe over Fabrics
NVMe/TCP	NVMe over TCP
ToR	Top of Rack
VE	Virtual Environment
VLAN	Virtual Local Area Network
VM	Virtual Machine

Revision history

Table 2. Revision history

Date	Document revision	Description of changes
February 2025	H04436	Initial version
September 2025	H04436.1	Updated with virtualization tasks

Code and file examples

NOTE: Many PDF viewers add a line break to the end of each line of text in a PDF. Adobe Acrobat (Reader, Standard, and Pro) and other common PDF viewers, including Google Chrome and Microsoft Edge, insert these line breaks. As a result, when you copy commands that wrap across multiple lines in a PDF, the copied commands contain erroneous line breaks. If you copy and paste commands, the line breaks cause issues.

To address this known limitation and ensure that copied commands do not contain unintentional line breaks, do one of the following:

- Paste the copied commands into a text editor and remove the line breaks.
- Use the HTML version of this document when you copy commands.

We value your feedback

Dell Technologies and the authors of this document welcome your feedback on the solution and the solution documentation. Contact the Dell Technologies Solutions team by email.

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NOTE: For links to additional documentation for this solution, see the Dell Technologies Solutions Info Hub for PowerStore.

Product overview

PowerStore

The PowerStore platform is a bare-metal unified storage array that can service block, file, and vVol resources along with numerous data services and efficiencies. PowerStore is perfect for traditional and modern workloads, including relational databases, electronic medical record applications, and content repositories among many others.

Dell PowerStore has two purpose-built platforms:

- The 2U, two-node, dual- socket Intel® Xeon® platform is used for the PowerStore 1000, 1200, 3000, 3200, 5000, 5200, 7000, 9000, and 9200 models.
- The 2U, two-node single-socket Intel® Xeon® platform is used for the PowerStore 500 model.

They run the exact same version of software and can be clustered and managed together coherently. For more information about PowerStore platforms, see Dell PowerStore: Introduction to the Platform.

Figure 1. PowerStore appliance

Organizations that use new applications and solutions require dependable storage, and often face the challenge of doing more with less. PowerStore addresses this challenge by packaging a powerful storage system into a cost-efficient and space-efficient profile. PowerStore features and benefits include:

- Active/active architecture
- Multiple protocol support including iSCSI, Fibre Channel, NVMe/FC, NVMe/TCP, SMB, FTP, SFTP, NFS, and vVol (with iSCSI, Fibre Channel and NVMe/FC)
- Unified offering serving block, file, and vVol storage
- Simplified, modern user interface
- VMware integration
- Inline data reduction
- Local and remote native data protection

PowerStore achieves new levels of operational simplicity and agility. It uses a container- based microservices architecture, advanced storage technologies, and integrated machine learning to unlock the power of your data. It offers predictive analytics to monitor, analyze, and troubleshoot the environment. PowerStore is highly adaptable, providing the flexibility to host specialized workloads and fits coherently into existing infrastructures to meet industry requirements. No matter where your business strategy takes you, the full-featured, adaptable design of PowerStore can help you build your unique breakthrough.

Proxmox Virtual Environment

Proxmox Virtual Environment (VE) is a robust open-source platform that is designed for server virtualization. It integrates two main virtualization technologies: Kernel-based Virtual Machine (KVM) for running virtual machines and Linux Containers (LXC) for lightweight container virtualization.

Key features of Proxmox Virtual Environment include:

Web-Based Management	A convenient web interface that enables you to manage VMs and containers, including deployment, monitoring, and configuration.
High Availability	Proxmox VE supports clustering, enabling high availability setups to ensure continuous operation and minimal downtime.
Backup and Restore	Integrated backup tools facilitate backup and restore processes for both VMs and containers.

Storage Options	It supports various storage solutions, including local storage, NFS, iSCSI, Ceph, and more, allowing flexible resource management.
Networking	Advanced networking features, such as virtual bridges and VLAN support, help in configuring complex network setups.
Open Source	As an open-source solution, Proxmox VE has a vibrant community for support and contributions, along with a subscription model for enterprise-level support.

NOTE: As Proxmox is based on Debian, it is a supported operating system on PowerStore. For more information, see Debian GNU/Linux Support Statement.

Solution architecture

Overview

This section describes the architecture of a Proxmox VE with PowerStore environment. In this solution, a Proxmox VE is deployed on three bare metal servers. Then, the networks that are required to access the PowerStore platform using block and file protocols are configured on the Proxmox VE nodes.

Logical architecture

The following diagram shows the logical architecture of this solution:

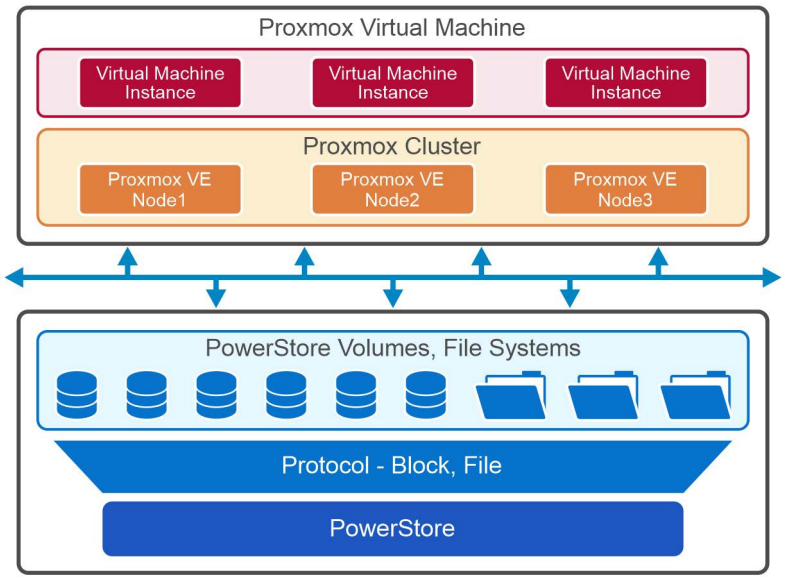

Figure 2. Logical architecture of Proxmox VE on Dell PowerStore

Dell PowerStore supports a unified (block and file) or block-optimized (block only) configuration. For this solution, a PowerStore 5200T single appliance cluster is configured as unified storage array. This array enables the host servers to access the storage using both the block and file protocols.

The host servers are deployed with Proxmox VE. In this solution, Proxmox VE nodes are further configured with iSCSI, NVMe/TCP, and NFS to access the PowerStore system. For more information about the configuration of host servers used for the Proxmox cluster, see Configuration details.

Network architecture

The following diagram shows the network architecture of this solution.

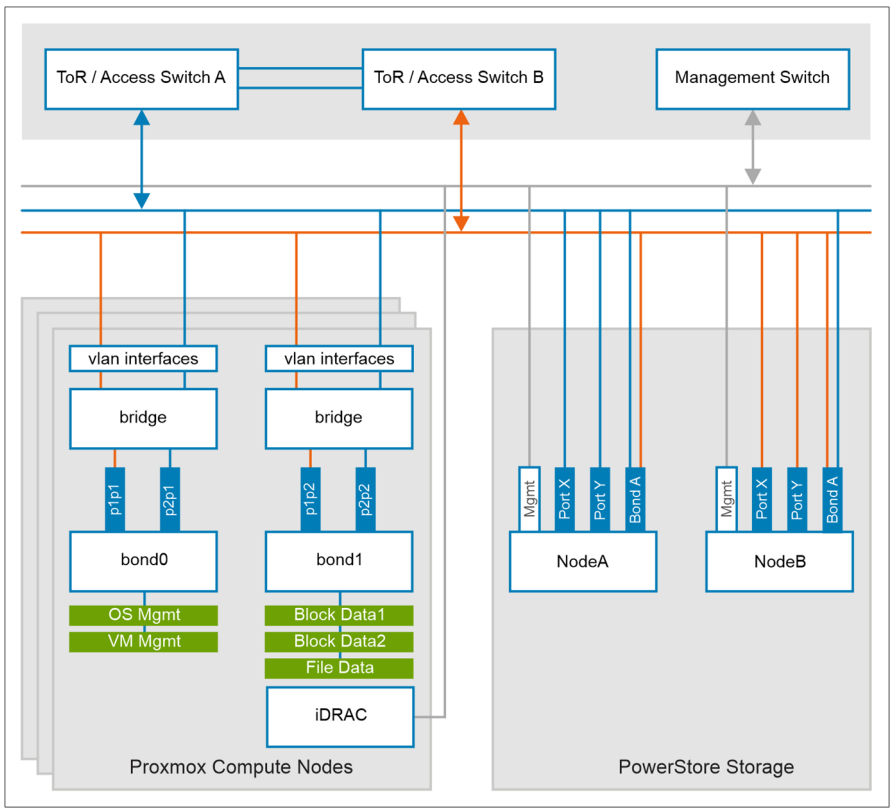

Figure 3. Proxmox VE with Dell PowerStore network design

At the physical layer, two supported Top of Rack (ToR) switches are used for redundancy and load-balancing purposes. A peer link is configured on both the ToR switches.

PowerStore supports Ethernet connectivity through ports on the embedded module, and on optional I/O modules. PowerStore supports Fibre Channel connectivity through ports on optional I/O modules.

In this example solution, compute hosts are connected to the PowerStore platform through Ethernet.

The network for storage services like iSCSI, NVMe/TCP, and NAS are configured on PowerStore. PowerStore appliances require all networks to be unique. Dell Technologies highly recommends that you deploy PowerStore networks with multiple and unique VLANs to separate the traffic.

For more information about the storage service networking and their configuration using the PowerStore Manager, see Networking Guide for Storage Services.

On the Proxmox nodes, two bond interfaces are created, each with two physical interfaces. Then, bridge interfaces are created on top of the bond interfaces. Finally, VLAN interfaces are created using the bridge device to separate different traffic types.

The following tables show the different networks that are configured for this solution.

Table 3. Proxmox host servers networks details

Network type	VLAN
Operating System (OS) Management	132
Virtual machine (VM) Management	133
Block Data 1 (iSCSI and NVMe/TCP)	1075
Block Data 2 (iSCSI and NVMe/TCP)	1076
File Data (NFS)	1077

Table 4. PowerStore storage networks details

Network type	VLAN	Description
PowerStore Cluster Management	132	**Mgmt**: PowerStore system management ports.
iSCSI	1075	**Port X**: iSCSI storage network that is mapped to ports on PowerStore.
NVMe/TCP	1076	**Port Y**: NVMe/TCP storage network that is mapped to ports on PowerStore.
File	1077	**Bond A**: The File network can be configured to use the system-created bond (bond0), a user-defined bond (link aggregation), or a fail-safe network.

Create a cluster in Proxmox

This section provides instructions for installing Proxmox VE on a host server using the Proxmox VE supplied ISO file.

Prerequisites

The following requirements must be met before installing Proxmox VE on a host server.

- Obtain the latest Proxmox installer ISO image from the Proxmox ISO download page.
- Obtain the DNS, default gateway, FQDN, and IP address details required for the host server configuration.

About this task

Complete the following procedure to install Proxmox VE.

Steps

1. Log in to the iDRAC using the host server iDRAC IP address.
2. Click **Launch Virtual Console** from the iDRAC.
3. Click **Connect Virtual Media**.
4. Click **Map CD/DVD** to map the downloaded Proxmox ISO image.
5. Click **Boot**.
6. Choose **Virtual CD/DVD/ISO**.
7. **Reboot** the host server.

   The Proxmox VE menu appears.

8. Select **Install Proxmox VE (Graphical)** to start the standard installation.

Figure 4. Proxmox installation menu

9. Read and accept the **EULA** to continue.
10. Choose the **Target Hard Disk** where Proxmox should be installed.

    NOTE: In this paper Proxmox VE is installed on the Dell BOSS card.

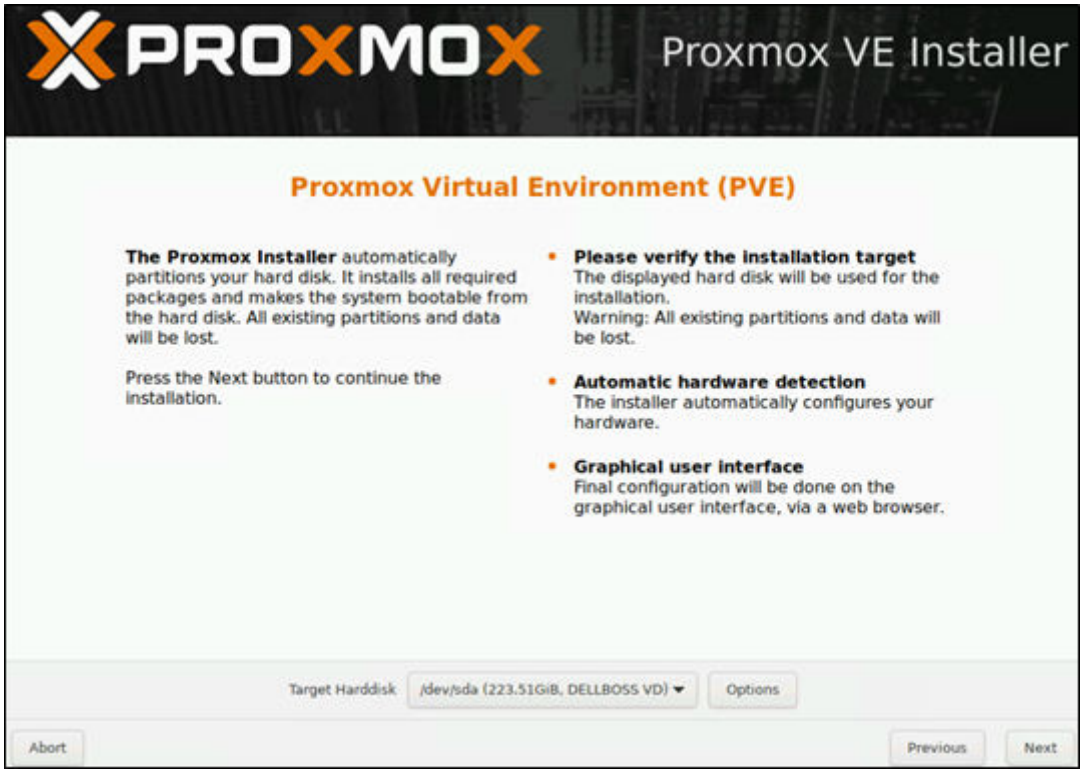

Figure 5. Target hard disk to install Proxmox

11. In the **Location and Time Zone** window:
    a. Update the:
        - Country
        - Time zone
        - Keyboard Layout
    b. Click **Next**.
12. Create a strong password for admin credentials.
    a. Reenter the password to confirm.
    b. Enter an email address for system administrator notifications.
13. Configure the Proxmox network configuration:
    a. Select the **Management Interface** and **Hostname** for the host server.
    b. Enter an available **IP Address**, **Gateway**, and **DNS Server**.
    c. Click **Next**.
14. Review the **Summary**.
    a. Check the **Automatically reboot after successful installation** checkbox.
    b. Click **Install** to start the installation.
15. Disconnect the Virtual Media after installation.

    Once the system reboots, the **Proxmox GRUB** menu loads.
16. Select **Proxmox VE GNU/Linux** and then click **Enter**.

    The Proxmox VE login prompt, along with a Welcome message and a URL to the Proxmox UI appears on the console.

Figure 6. Proxmox VE login prompt

17. Go to the URL using a web browser.
18. To access the interface:
    a. Log in as root.
    b. Provide the **Password** set during the Proxmox VE installation.

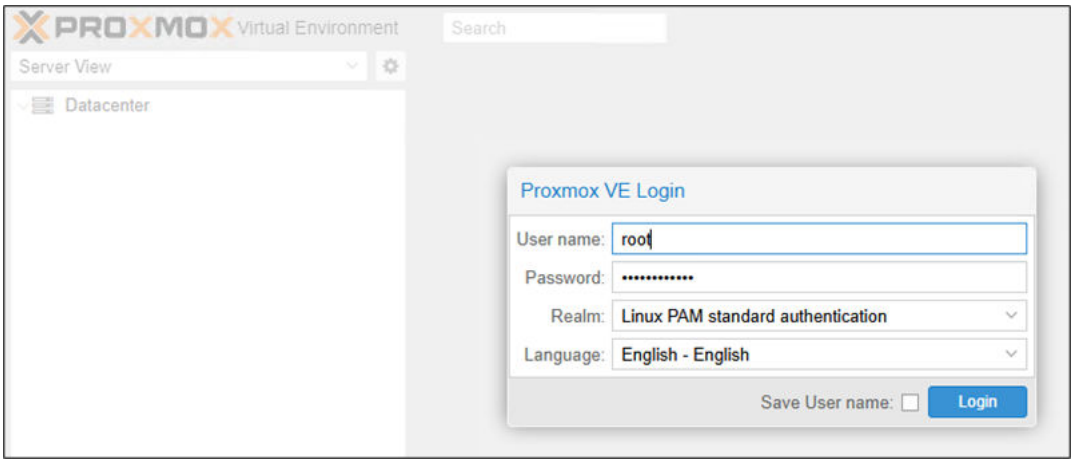

Figure 7. Proxmox UI login screen

NOTE: Proxmox VE requires a minimum of three nodes to create a cluster.

19. Repeat the steps above to install Proxmox VE on the other two host servers.
20. Configure time synchronization on each Proxmox host server using **Chrony**.

    For more information about the `chrony` configurations, see Proxmox Time Synchronization.

    For more information about the Proxmox installation on host servers, see Installing Proxmox VE.

Network configuration in Proxmox VE

Configuration

Proxmox VE uses the Linux network stack. The network configuration can be done either using the UI, or by manually editing the `/etc/network/interfaces` file.

In this solution, the network is configured using bond, bridges, and VLAN interfaces as described in Network architecture.

- A Linux bond interface is used to bind multiple NICs to a single network device providing fault tolerance and load balancing.
- A Linux bridge interface (commonly called vmbrX) is used to connect guests to the underlying physical network.
- A Linux VLAN interface is used for network segmentation on the same physical network interface. Proxmox supports 802.1q VLAN tagging, which can be configured on bridges to enable VLAN support for VMs.

For an example `/etc/network/interfaces` file, see Proxmox network configuration file.

For more information about Proxmox network configuration, see Network Configuration.

NOTE: There is a 16-character length limit on the network interface names. If the network interface name exceeds 16 characters, then network configuration fails. To fix this issue, override the network interface name. For more information, see Override network device names.

Create a cluster in Proxmox VE

This section provides instructions for creating a Proxmox cluster using three host servers to provide high availability, load balancing, and centralized web management.

Prerequisites

The following requirements must be met before creating a Proxmox cluster:

- All host servers are installed with Proxmox VE.
- The date and time must be synchronized on all host servers.

About this task

Complete the following procedure to configure a cluster.

Steps

1. Log in to the Proxmox UI on the first host server.
   a. Click **Datacenter** > **Cluster** .
   b. In the **Cluster Information window**, click **Create Cluster**.
   c. Provide a **Name** for the cluster.
   d. Choose a network interface for the **Cluster Network** and then click **Create**.

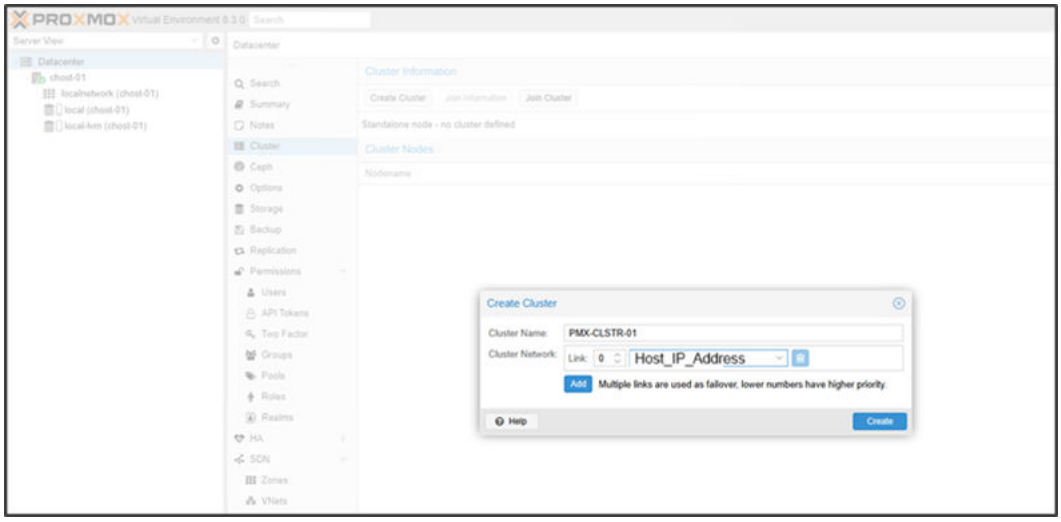

Figure 8. Create cluster

2. Once the cluster is created:
   a. Click **Join Information**.
   b. Click **Copy Information** to copy the **Join Information**.

   Other nodes use this information while joining this cluster.

3. Log in to the Proxmox UI of the second host server that must be added to the cluster.
4. To join the Cluster:
   a. Click the **Cluster** tab.
   b. Select **Join Cluster**.
   c. Provide the **join information**, which was copied from the first node.
   d. Enter the root **Password** of the first host server.
   e. Select the network interface on this node which is used for the **Cluster Network**.
   f. Click Join **<cluster_name>**.

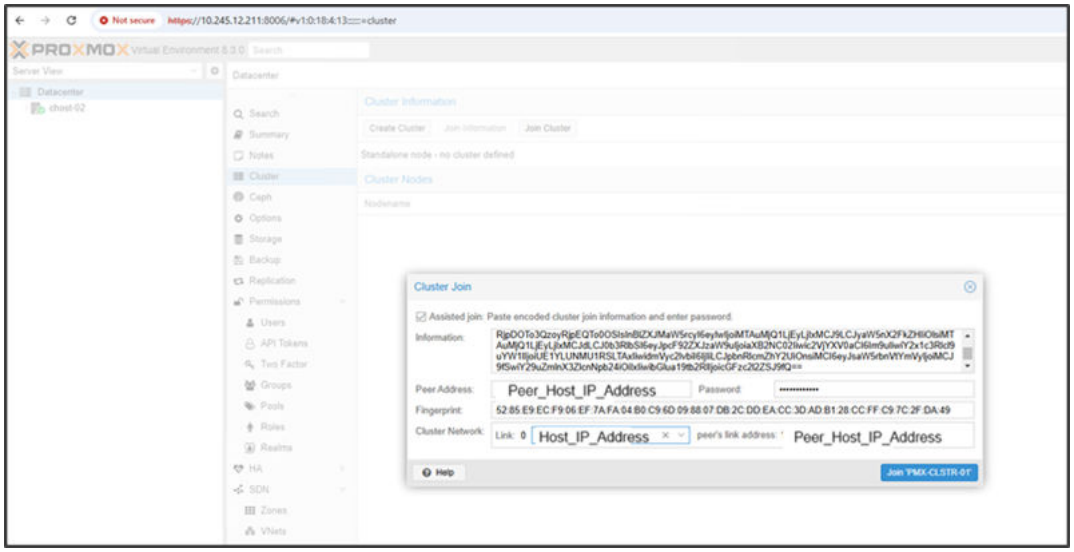

Figure 9. Cluster join

5. Repeat the above steps to join the third host server to the cluster.
6. After the process is complete, login to the Proxmox UI of the first node.
   a. Check the **Cluster** tab to view the hosts which are part of this cluster.

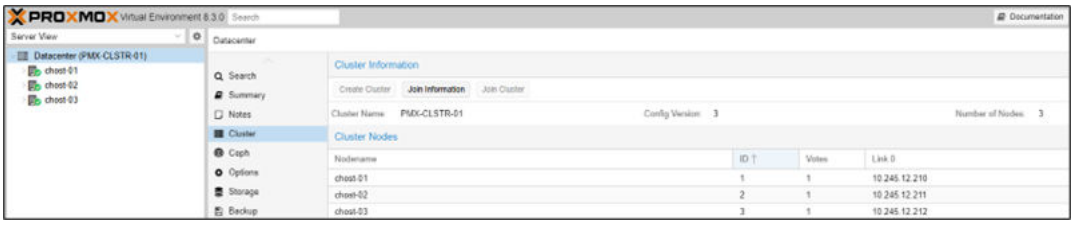

Figure 10. Cluster Information

For more information about creating a cluster, see Create a cluster.

iSCSI connectivity

Overview

NOTE: This section is required only if iSCSI is used for PowerStore volume connectivity.

This section provides instructions to:

- Configure the iSCSI initiator on the Proxmox VE nodes.
- Discover and connect to the PowerStore iSCSI target.
- Multipath configuration on the iSCSI LUN in the host.
- LVM configuration on the iSCSI multipath device.

Prerequisites

The following requirements must be met before configuring iSCSI on Proxmox VE nodes:

- The network to access the iSCSI target should be configured on the Proxmox host servers. For more information about configuring the network on Proxmox VE nodes, see Network configuration in Proxmox VE.
- The iSCSI network should be configured on the PowerStore platform. For more information about configuring the iSCSI storage network on PowerStore, see Create Block-optimized Storage networks in PowerStore Manager.

iSCSI configuration

About this task

Follow the below steps to configure the iSCSI initiator, discover, and connect to iSCSI target.

Steps

1. On the Proxmox VE nodes, install the `open-iscsi` package.

```
# apt-get install open-iscsi
```

2. Check the iSCSI initiator iSCSI Qualified Name (IQN) and make a note of it.

```
# cat /etc/iscsi/initiatorname.iscsi
InitiatorName=iqn.1993-08.org.debian:01:db789a9e9034
```

3. In the Proxmox UI, go to **Datacenter** > **Storage** > **Add** > **iSCSI**.

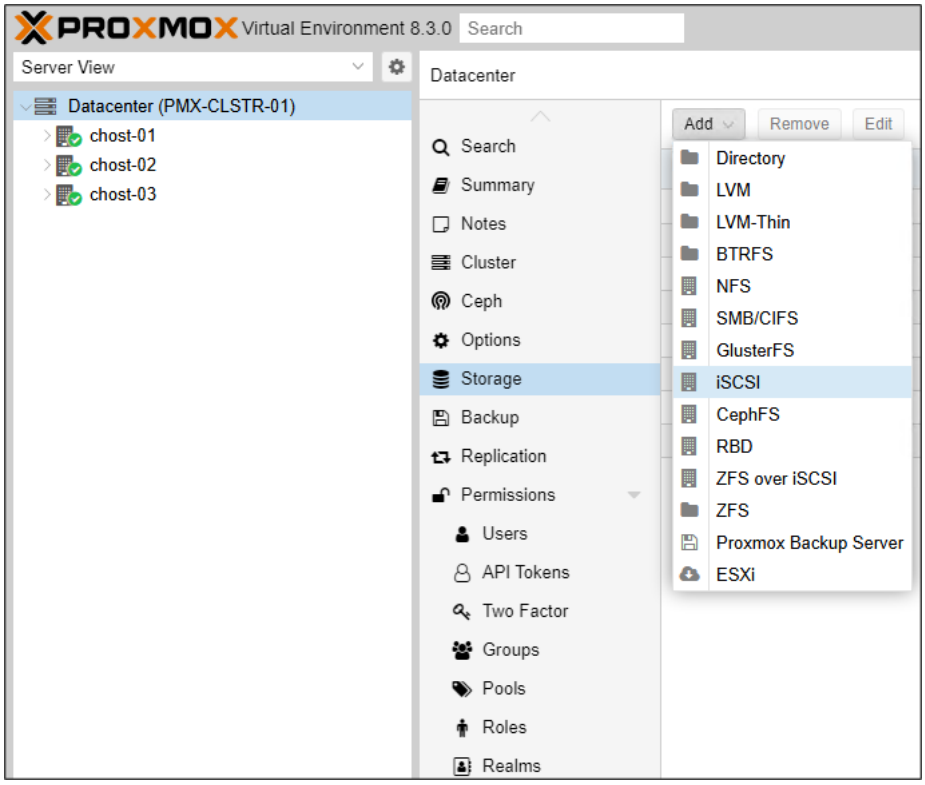

Figure 11. iSCSI configuration

4. Provide a name in the **ID** field.
   a. Provide the iSCSI target IP configured on PowerStore in the **Portal** field.

      The target IQN is automatically discovered.

   b. Choose the required target **IQN** to connect from the drop-down list.
   c. In the **Nodes** field, select all the nodes of the cluster.
   d. Select the **Enable** check box.
   e. Clear the **Use LUNs directly** check-box.
   f. Click **Add**.

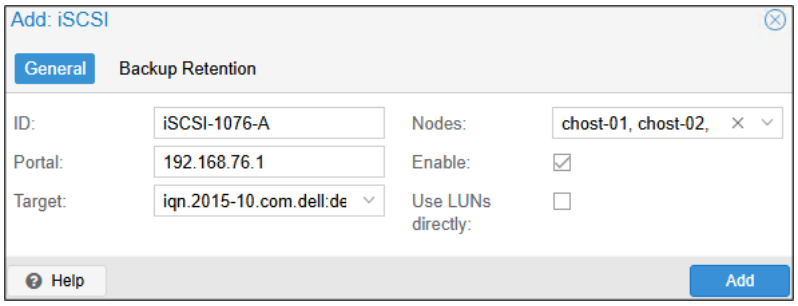

Figure 12. iSCSI target discovery and login

5. Repeat the steps above to discover all the iSCSI target ports from the iSCSI data networks configured on the host.

   In this solution, two iSCSI data networks are configured per host. Four iSCSI target networks are configured on the PowerStore platform.

6. In PowerStore Manager, add the host by going to **Compute** > **Host Information** and clicking **+ Add Host**.
7. In the **Add Host** window:
   a. Provide a **Host Friendly Name**.
   b. Select **Linux** as the **Operating System** from the drop-down menu.

c. Click **Next**.

8. Select **iSCSI** as the **Initiator Type** and click **Next**.

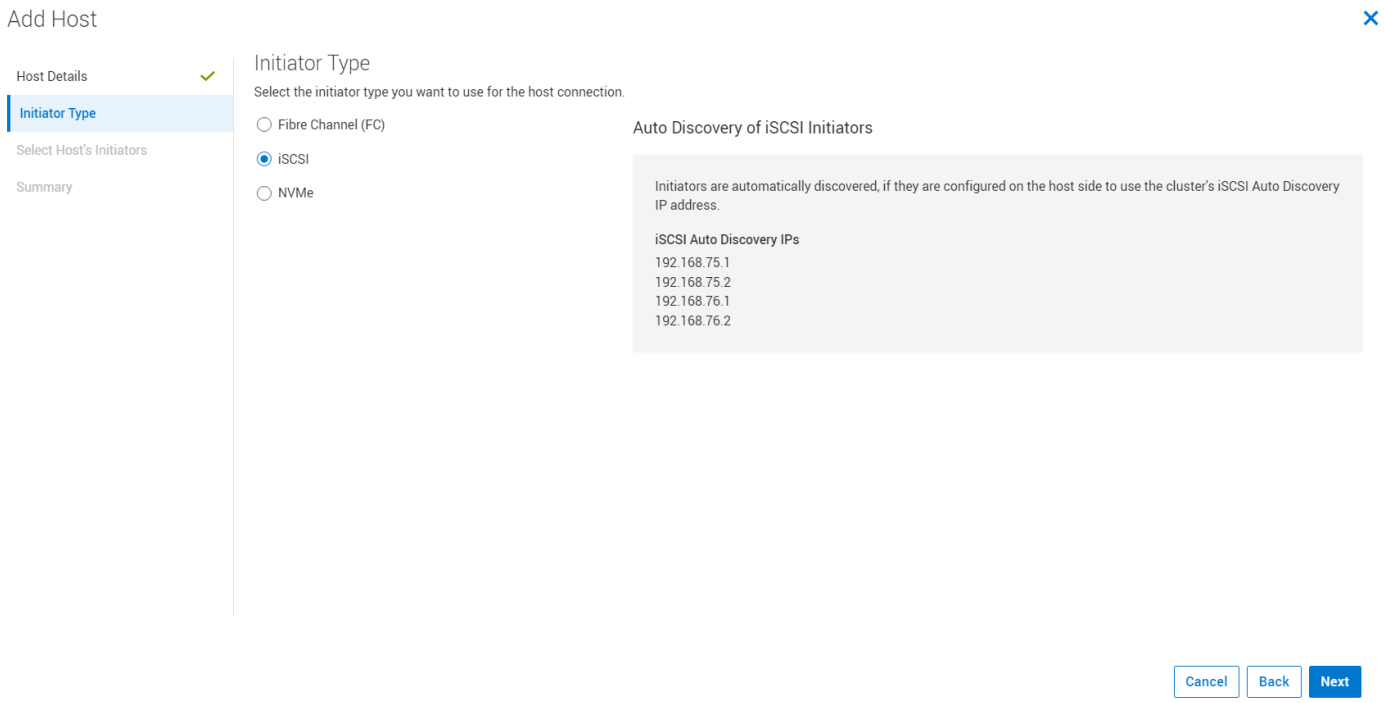

Figure 13. iSCSI host discovery

The host is automatically discovered.

9. Choose the host **IQN** and then click **Next**.

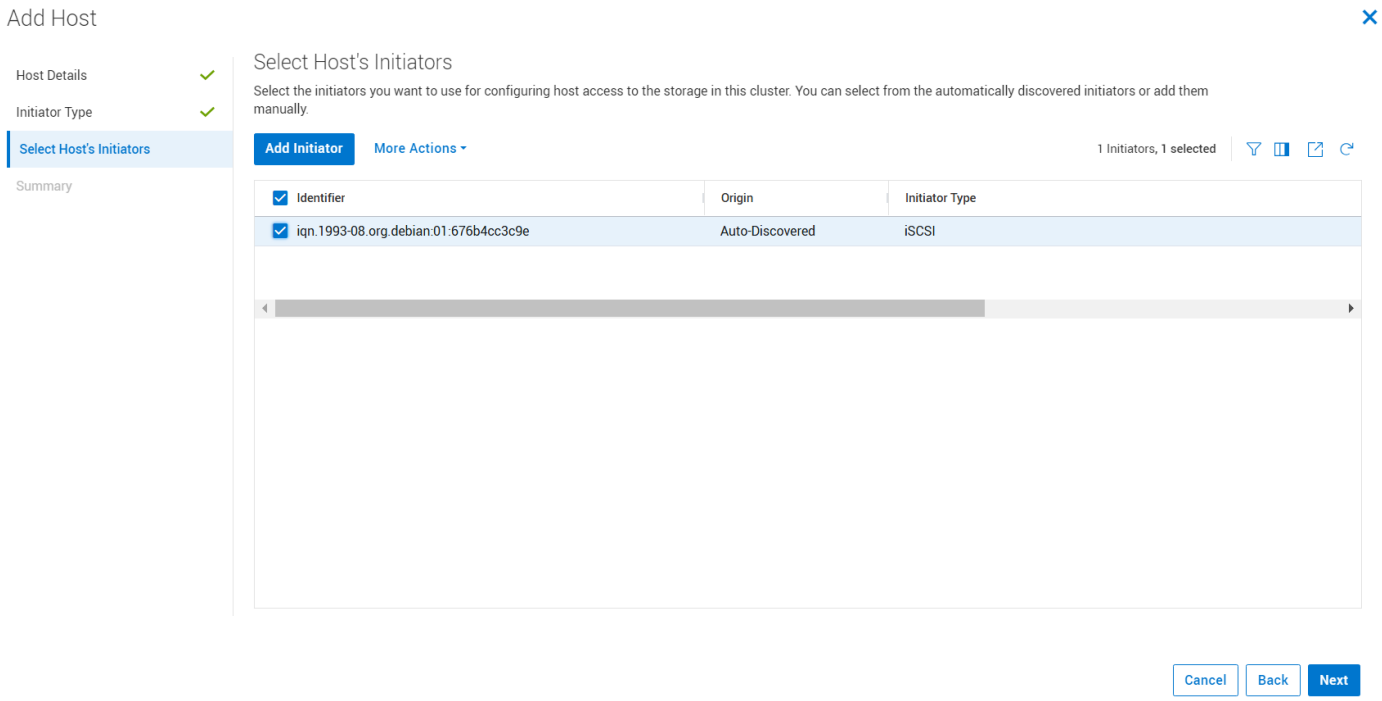

Figure 14. Automatically discovered iSCSI hosts

10. Select **Local Connectivity** in the **Host Connectivity** window and then click **Next**.
11. Review the **Summary** and then click **Add Host**.

12. Repeat the steps above in PowerStore Manager to add all three Proxmox hosts.
13. In PowerStore Manager, create a volume by going to **Storage** > **Volumes** .
    a. Click **+ Create**.
14. In the **Properties** window:
    a. Provide a **Name** for the volume.
    b. Choose the **Category** and the **Application** type that will use this volume.
    c. Provide the required **Size** of the volume.
    d. Click **Next**.
15. In the **Host Mappings** window:
    a. Select **SCSI** as the protocol type to view all the iSCSI hosts configured on PowerStore.
    b. Select the iSCSI hosts to which the volume should be mapped and then click **Next**.

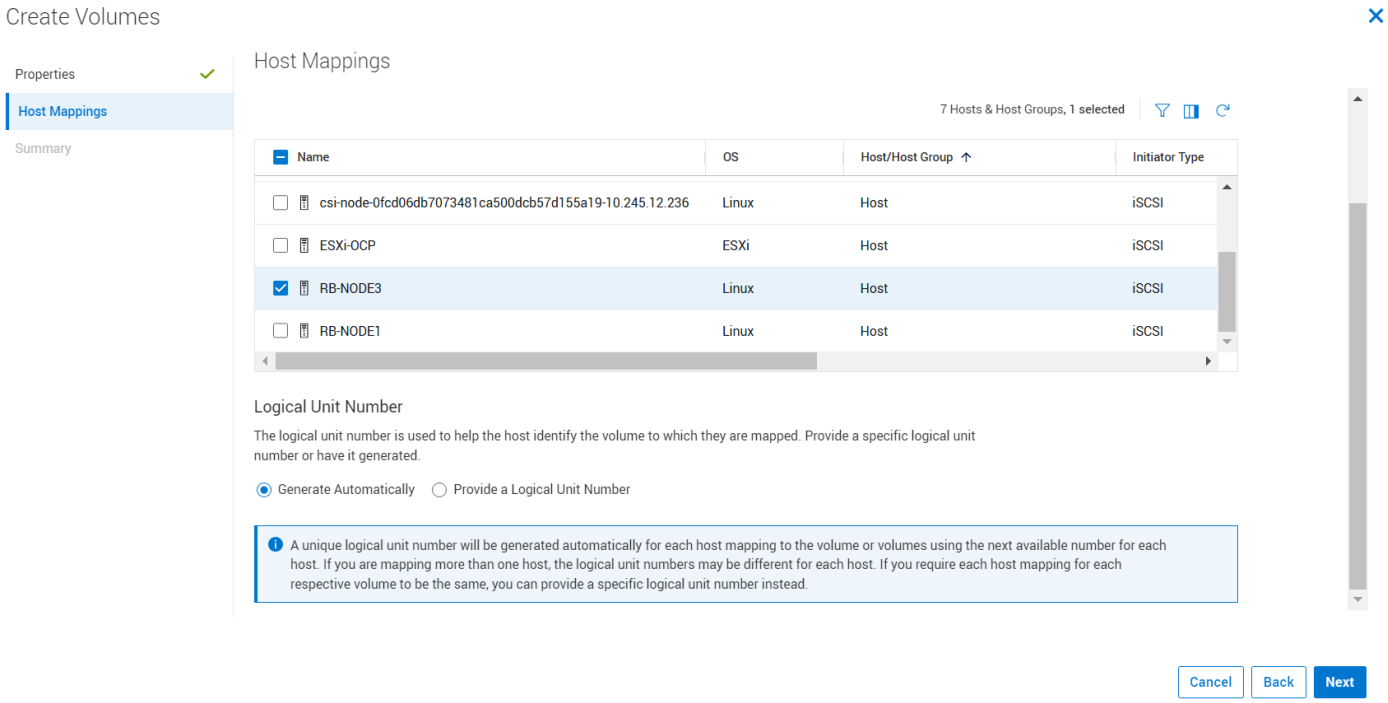

Figure 15. Creating and mapping volume to iSCSI hosts

16. Review the **Summary** and then click **Create**.

    This action creates a volume and maps it to the Proxmox hosts.

Multipath configuration

This topic describes the procedure to configure multipath.

About this task

The volume that is mapped in iSCSI configuration appears as multiple block devices on the Proxmox host.

```
root@chost-02:~# lsblk
NAME                MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda                   8:0    0 223.6G  0 disk
├─sda1                8:1    0  1007K  0 part
├─sda2                8:2    0     1G  0 part /boot/efi
└─sda3                8:3    0 222.6G  0 part
  ├─pve-swap        252:0    0     8G  0 lvm  [SWAP]
  ├─pve-root        252:1    0  65.6G  0 lvm  /
  ├─pve-data_tmeta  252:2    0   1.3G  0 lvm
  │ └─pve-data      252:4    0 130.3G  0 lvm
  └─pve-data_tdata  252:3    0 130.3G  0 lvm
    └─pve-data      252:4    0 130.3G  0 lvm
sdb                   8:16   0   160G  0 disk
sdc                   8:32   0   160G  0 disk
sdd                   8:48   0   160G  0 disk
sde                   8:64   0   160G  0 disk
root@chost-02:~#
```

Figure 16. Volume seen as multiple block devices

The volume appears so because of the multiple paths through which the host is connected to the volume. Each block device corresponds to one particular path. If that path fails, the block device is unresponsive. Multipath should be configured in order to obtain fault tolerance and load balancing.

Follow the instructions below to configure multipath.

Steps

1. Install the `multipath-tools` package:

```
# apt update
# apt install multipath-tools
```

2. Use the `scsi_id` command to query the WWID of a specific device.

   The following command shows the WWID of device /dev/sdb:

```
# /lib/udev/scsi_id -g -u -d /dev/sdb
368ccf09800886405511a87f3516fc901
```

3. Run the following command to add the WWIDs of all LUNs with multipath connectivity to the /etc/multipath/wwids file:

```
# multipath -a 368ccf09800886405511a87f3516fc901
```

4. Run the following command to activate the settings:

```
# multipath -r
```

5. Run the following command to view the multipath device topology:

```
# multipath -ll
```

```
root@chost-02:~# multipath -ll
368ccf09800886405511a87f3516fc901 dm-5 DellEMC,PowerStore
size=160G features='1 queue_if_no_path' hwhandler='1 alua' wp=rw
|-+- policy='service-time 0' prio=50 status=active
| |- 16:0:0:1 sdb 8:16 active ready running
| `- 17:0:0:1 sdc 8:32 active ready running
`-+- policy='service-time 0' prio=10 status=enabled
  |- 18:0:0:1 sdd 8:48 active ready running
  `- 19:0:0:1 sde 8:64 active ready running
root@chost-02:~#
```

Figure 17. Multipath device topology

6. Create a multipath configuration file `/etc/multipath.conf`, to include device-specific information.

A sample `/etc/multipath.conf file` is provided below:

```
# cat /etc/multipath.conf
defaults {
        polling_interval 5
        checker_timeout 15
        disable_changed_wwids yes
        find_multipaths no
        user_friendly_names yes
        enable_foreign nvme
}
devices {
        device {
                vendor DellEMC
                product PowerStore
                detect_prio "yes"
                path_selector "queue-length 0"
                path_grouping_policy "group_by_prio"
                path_checker tur
                failback immediate
                fast_io_fail_tmo 5
                no_path_retry 3
                rr_min_io_rq 1
                max_sectors_kb 1024
                dev_loss_tmo 10
                hardware_handler "1 alua"
        }
}
```

7. Restart the `multipathd` service using the following command:

```
# systemctl restart multipathd.service
```

LVM on an iSCSI multipath device configuration

About this task

The LVM should be configured on the multipath device in order to use the iSCSI multipath device as storage for virtual machines in the Proxmox environment. Follow the instructions below to create the LVM.

Steps

1. Create a Physical Volume (PV) and Volume Group (VG) on top of the multipath device by running the following commands:

```
root@chost-02:~# pvcreate /dev/mapper/mpathb
Physical volume "/dev/mapper/mpathb" successfully created.

root@chost-02:~# vgcreate vgs1 /dev/mapper/mpathb
Volume group "vgs1" successfully created
```

2. In the Proxmox UI:
   a. Go to **Datacenter** > **Storage** > **Add** > **LVM** to create LVM storage for the VG created above.
3. In the **Add: LVM** window
   a. Provide a name for the LVM storage in the **ID** field.
   b. In the drop-down menu of the **Volume group** field, select the VG created above.

      The figure below shows the VGs configured on the host.

   c. Select the correct host in the **Node to scan** field.
   d. Click **Add**.

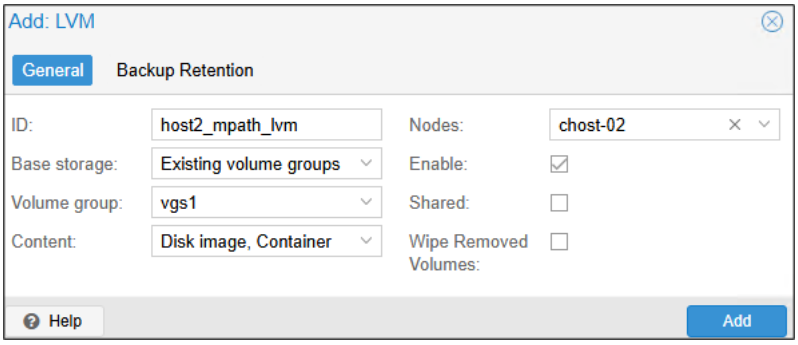

Figure 18. LVM on the iSCSI multipath device

Results

The LVM storage that was created can be used while configuring storage for the virtual machines.

LVM-thin configuration

About this task

Proxmox permits the creation of LVM or LVM-thin on block storage like iSCSI or NVMe/TCP LUNs. With LVM-thin, blocks are allocated on the storage as they are written. This behavior is called thin provisioning.

LVM backend can be used with shared storage whereas LVM-thin backend can be used only with non-shared storage. Also, LVM-thin supports VM snapshots and linked clones.

Follow the instructions below to create LVM-thin on an iSCSI multipath device. In the below example, an iSCSI LUN of 112 GB from PowerStore is mapped to the host. This iSCSI LUN is detected as /dev/mapper/mpathe on the host (rb-pve2):

Steps

1. Create a Physical Volume (PV) and Volume Group (VG) on top of the multipath device by running the following commands:

```
root@rb-pve2:~# pvcreate /dev/mapper/mpathe
  Physical volume "/dev/mapper/mpathe" successfully created.
root@rb-pve2:~# vgcreate vgs2 /dev/mapper/mpathe
  Volume group "vgs2" successfully created
```

2. Create an LVM-thin pool on the new VG - vgs2, by running the following command:

```
root@rb-pve2:~# lvcreate -L 110g -T vgs2/LT_iSCSI
  Thin pool volume with chunk size 64.00 KiB can address at most <15.88 TiB of data.
  Logical volume "LT_iSCSI" created.
```

3. To create LVM-thin storage using the above created pool, in the Proxmox UI go to **Datacenter** > **Storage** > **Add** > **LVM-Thin**.
4. In the **Add: LVM-Thin** window, provide a **name** for the LVM-thin storage in the **ID** field.
5. In the drop-down menu of **Volume group** field, select the correct host in the **Node to scan** field.
6. This shows the VGs configured on the host.
7. Select the **VG** created in the above step and click **Add**.

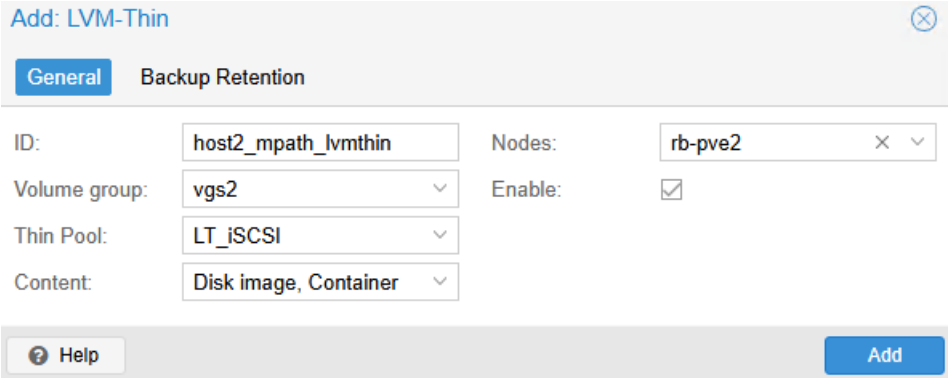

Figure 19. LVM-thin on iSCSI multipath

The LVM-thin storage created can be used while configuring storage for the virtual machines.

NOTE: For more information about the disk formats and storage types supported by Proxmox, see Storage Types.

NVMe over TCP connectivity

Overview

NOTE: This section is required only if NVMe/TCP is used for PowerStore volume connectivity.

This section provides instructions to:

- Configure the NVMe/TCP initiator on the Proxmox VE nodes.
- Discover and connect to the PowerStore NVMe/TCP target.
- Configure multipath on the NVMe/TCP LUN in the host.
- Configure LVM on the iSCSI multipath device.

Prerequisites

The following requirements must be met before configuring NVMe/TCP on Proxmox VE nodes:

- The network to access the NVMe/TCP target should be configured on the Proxmox host servers. For more information about configuring the network on Proxmox VE nodes, see Network configuration in Proxmox VE.
- The NVMe/TCP network should be configured on the PowerStore platform. For more information about configuring the NVMe/TCP storage network on PowerStore, see Create Block-optimized Storage networks in PowerStore Manager.

NVMe/TCP configuration

About this task

Follow the steps below to configure the NVMe/TCP initiator, and to discover and connect to the NVMe/TCP target.

Steps

1. On the Proxmox nodes, install the `nvme-cli` package:

```
# apt update && apt -y install nvme-cli
```

2. Load the `nvme_tcp` driver and check if the driver is loaded.

```
# modprobe nvme_tcp
# lsmod | grep nvme
nvme_tcp               53248  0
nvme_keyring           16384  1 nvme_tcp
nvme_fc                49152  1 qla2xxx
nvme_fabrics           36864  2 nvme_tcp,nvme_fc
nvme_core             204800  3 nvme_tcp,nvme_fc,nvme_fabrics
nvme_auth              24576  1 nvme_core
```

3. Check the host NVMe Qualified Name (NQN) and make a note of it.

```
# cat /etc/nvme/hostnqn
nqn.2014-08.org.nvmexpress:uuid:4c4c4544-0050-5a10-8047-c8c04f315332
```

4. To configure the `nvme_tcp` driver to be loaded at every boot, run the following command:

```
# echo "nvme_tcp" > /etc/modules-load.d/nvme_tcp.conf
```

5. Run the following command to discover the NVMe/TCP target:

```
# nvme discover -t tcp -a <NVMe/TCP target IP address>
```

6. Run the following command to connect to the discovered NVMe/TCP target:

```
# nvme connect -t tcp -n <NQN of the NVMe/TCP target> -a <NVMe/TCP target IP address>
```

7. Repeat the step above to discover all the NVMe/TCP target ports from the data networks configured on the host.

   In this solution, two data networks are configured per host. Four NVMe/TCP target networks are configured on the PowerStore platform.

8. In PowerStore Manager: add the host by going to **Compute** > **Host Information**.
   a. Click **+ Add Host**.
9. In the **Add Host** window:
   a. Provide a **Host Friendly Name**.
   b. Select **Linux** as the **Operating System** from the drop-down menu.
   c. Click **Next**.
10. Select **NVMe** as the **Initiator Type** and then click **Next**.

    The host is automatically discovered.

11. Choose the host **NQN** and then click **Next**.
12. Review the **Summary** and then click **Add Host**.
13. Repeat the steps above in PowerStore Manager to add all the three Proxmox hosts.
14. In PowerStore Manager, create a volume by going to **Storage** > **Volumes**.
    a. Click **+ Create**.
15. In the **Properties** window:
    a. Provide a **Name** for the volume.
    b. Choose the **Category** and the **Application** type that will use this volume.
    c. Provide the required **Size** of the volume.
    d. Click **Next**.
16. In the **Host Mappings** window:
    a. Select **NVMe** as the protocol type to view all the NVMe hosts configured on PowerStore.
    b. Select the NVMe hosts to which the volume should be mapped.
    c. Click **Next**.
17. Review the **Summary** and then click **Create**.

    This action creates a volume and maps it to the Proxmox hosts.

Multipath configuration

About this task

In Proxmox, NVMe native multipath is the default for configuring multipath on NVMe devices.

Steps

1. To verify the NVMe multipath status, run the following command:

```
# cat /sys/module/nvme_core/parameters/multipath
Y
```

NOTE: The `Y` indicates that NVMe native multipath is enabled.

2. To view the NVMe multipath device, run the following command:

```
# nvme show-topology
```

```
root@chost-02:~# nvme show-topology
nvme-subsys1 - NQN=nqn.1988-11.com.dell:powerstore:00:8a9e82dd3e03EA879BDB
\
 +- ns 70
 \
  +- nvme1 tcp traddr=192.168.75.2,trsvcid=4420,src_addr=192.168.75.211 live non-optimized
  +- nvme2 tcp traddr=192.168.75.1,trsvcid=4420,src_addr=192.168.75.211 live optimized
  +- nvme3 tcp traddr=192.168.76.2,trsvcid=4420,src_addr=192.168.76.211 live optimized
  +- nvme4 tcp traddr=192.168.76.1,trsvcid=4420,src_addr=192.168.76.211 live non-optimized
root@chost-02:~#
```

Figure 20. NVMe multipath device topology

3. For the NVMe/TCP device to persist across reboots, create an entry in the `/etc/nvme/discovery.conf` file as shown below:

```
# echo "discover -t tcp -a 192.168.76.2 -s 4420" | tee -a /etc/nvme/discovery.conf
```

   a. Repeat the above command with all the target port IP addresses.

4. Enable the `nvmf-autoconnect` service to start upon boot on each Proxmox host by running the following command:

```
# systemctl enable nvmf-autoconnect.service
```

5. To ensure that NVMe pseudo device names are consistent across host reboots, configure `udev` rules as shown below:

```
# cat /etc/udev/rules.d/98-nvme.rules
KERNEL=="nvme*[0-9]n*[0-9]", SUBSYSTEM=="block", ENV{DEVTYPE}=="disk", SYMLINK+="nvme-
$attr{wwid}"
```

6. Reboot the host to ensure that the `udev` rule takes effect.

   Under `/dev`, there is a new device name that is associated with a World Wide Identifier (WWID). Dell Technologies recommends that you use this device name.

```
root@chost-02:~# ls -l /dev/nvme-*
lrwxrwxrwx 1 root root       7 Jan 16 11:45 /dev/nvme-eui.0ffb5cdb6f316a3d8ccf096800f670a4 -> nvme1n1
crw------- 1 root root 10, 122 Jan 16 09:32 /dev/nvme-fabrics
root@chost-02:~#
```

Figure 21. Persistent NVMe device name

LVM on a NVMe multipath device

About this task

The LVM should be configured on the multipath device in order to use the NVMe multipath device as storage for virtual machines in the Proxmox environment. Follow the instructions below to create the LVM.

Steps

1. Create a Physical Volume (PV) and Volume Group (VG) on top of the multipath device by running the following commands:

```
root@chost-02:~# pvcreate /dev/nvme-eui.0ffb5cdb6f316a3d8ccf096800f670a4
  Physical volume "/dev/nvme-eui.0ffb5cdb6f316a3d8ccf096800f670a4" successfully
created.

root@chost-02:~# vgcreate nvg1 /dev/nvme-eui.0ffb5cdb6f316a3d8ccf096800f670a4
  Volume group "nvg1" successfully created
```

2. To create LVM storage for the VG created above, go to **Datacenter** > **Storage** > **Add** > **LVM** in the Proxmox UI.
3. In the **Add: LVM** window:
   a. Provide a name for the LVM storage in the **ID** field.
   b. In the drop-down menu of the **Volume group** field, select the VG created in the step above.

      The figure below shows the VGs that are configured on the host.

   c. Select the correct host in the **Node to scan** field, and then click **Add**.

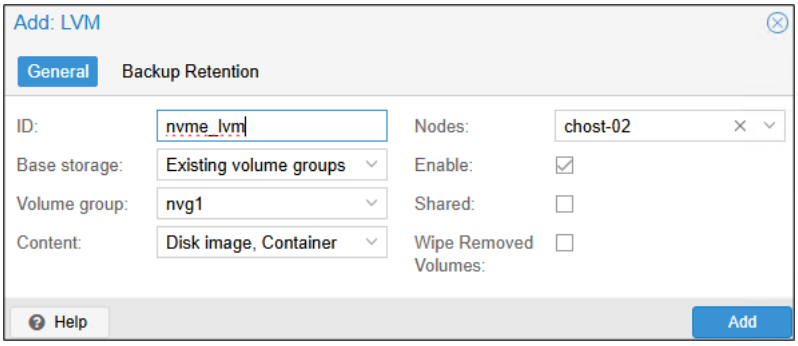

Figure 22. LVM on the NVMe multipath device

Results

The LVM storage that is created can be used while configuring storage for the virtual machines.

NFS connectivity

Overview

NOTE: The following section is required only if NFS is used for PowerStore file connectivity.

The following section provides instructions to configure NFS storage on the Proxmox cluster.

Prerequisites

The following requirements must be met before configuring NFS on Proxmox VE nodes:

- The network to access the NFS target should be configured on the Proxmox host servers. For more information about configuring the network on Proxmox VE nodes, see Network configuration in Proxmox VE.
- NFS should be configured on the PowerStore platform. For more information, see Dell PowerStore Configuring NFS.

NFS configuration on the host

About this task

Follow the below steps to add NFS storage on the Proxmox cluster.

Steps

1. In the Proxmox UI, go to **Datacenter** > **Storage** > **Add** > **NFS**.
2. In the **Add: NFS** window:
   a. Provide a name in the **ID** field.
   b. Provide the PowerStore **NFS IP address**.
   c. Choose the **NFS directory** from the drop-down menu of the **Export** field.
   d. Choose the kind of content that the NFS storage should allow from the **Content** drop-down menu.
   e. In the **Nodes** field, choose all the nodes of the cluster.
   f. Click **Add**.

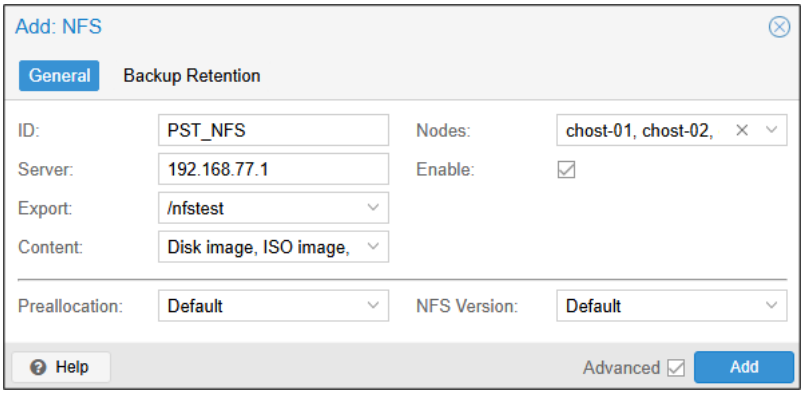

Figure 23. Adding NFS storage to a host

The NFS storage is now mounted on all the nodes of the cluster. In the UI, NFS storage is visible under each of the hosts.

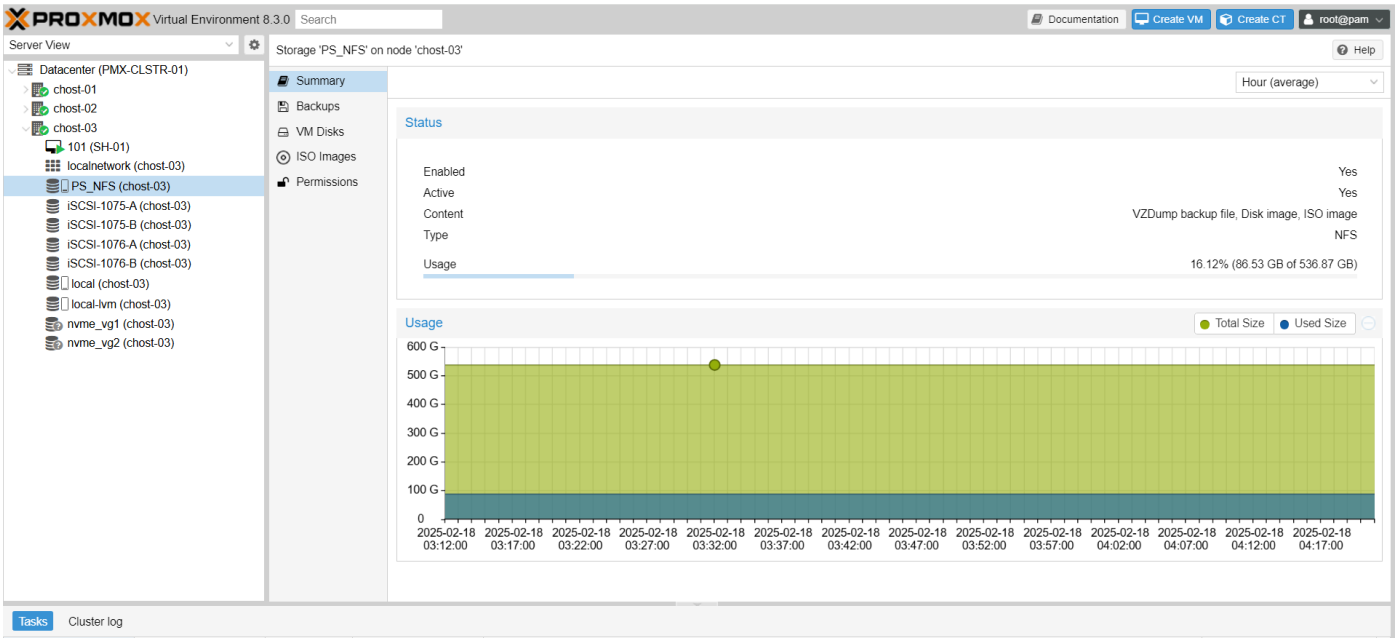

Figure 24. NFS storage added to a host

3. Click the **NFS storage**.
   a. Select the ISO image.
   b. Click **Upload** to upload an ISO image.

Virtual Machines

This section describes Virtual Machine (VM) deployment, the various operations that can be performed on VMs like templates, clones, VM migration, storage migration, and snapshots.

Create a virtual machine in Proxmox VE

This section provides instructions for creating the virtual machine in Proxmox VE.

Prerequisites

The following requirements must be met before creating a virtual machine in the Proxmox VE:

- The operating system ISO images should be available in the NFS share. See NFS connectivity for details on creating NFS share and uploading ISO.
- Two iSCSI disks from the PowerStore platform should be mapped to the Proxmox host and LVM should be created on these iSCSI disks. See iSCSI connectivity for details on configuring Proxmox hosts to access iSCSI disks and creating LVM on the iSCSI disks.

About this task

This procedure provides instructions to deploy a Virtual Machine (VM) in the Proxmox cluster. In this example solution, the VM is configured with iSCSI data disks and installed using an ISO available on the NFS share. The PowerStore platform provides both the iSCSI disks and NFS share.

Follow these steps to create a virtual machine in Proxmox VE.

Steps

1. Log in to the Proxmox VE UI.
2. To create a virtual machine, click **Datacenter** > **Create VM**.

Figure 25. Create VM

3. In the **Create: Virtual Machine** window:
   a. In the **General** tab:
      i. Select the **Node** on which the VM should be deployed.
      ii. Provide an **ID** and **Name** for the VM.
      iii. Click **Next**.
   b. In the **OS** tab:
      i. Select the NFS share from the drop-down menu of the **Storage** field.
      ii. Choose the ISO from the drop-down menu of the **ISO image** field.
      iii. Choose **Guest OS Type** and its **Version**.
      iv. Click **Next**.

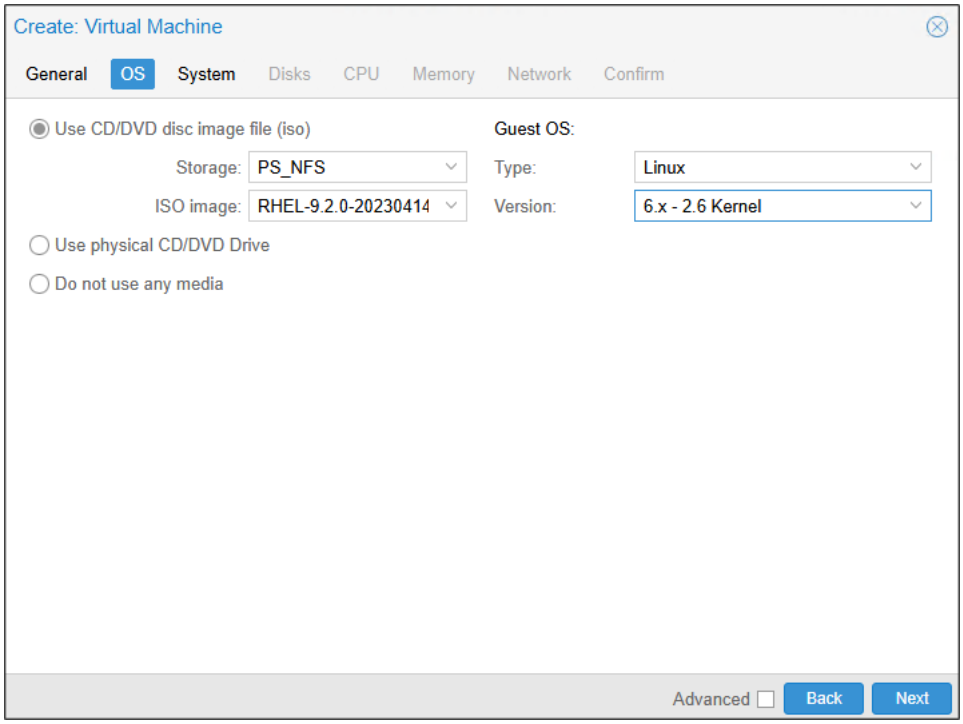

Figure 26. OS image from the NFS share

c. In the **System** tab:
   i. Configure the following as required:
      - **Graphic card**
      - **Machine type**
      - **SCSI controller**
      - **BIOS**
   ii. Click **Next**.

In the **Disks** tab, `scsi0` identifies the disk on which the operating system will be installed. In this solution, the operating system is installed on the `local-lvm` (local-disk) of the host.

4. To add additional disks, click **+ Add**.
   a. Select the LVM or the LVM-thin disk configured on the iSCSI multipath disk from the **Storage** drop-down menu.
   b. Provide a **Disk size**.
   c. Repeat the same process to add the second iSCSI disk to the VM.

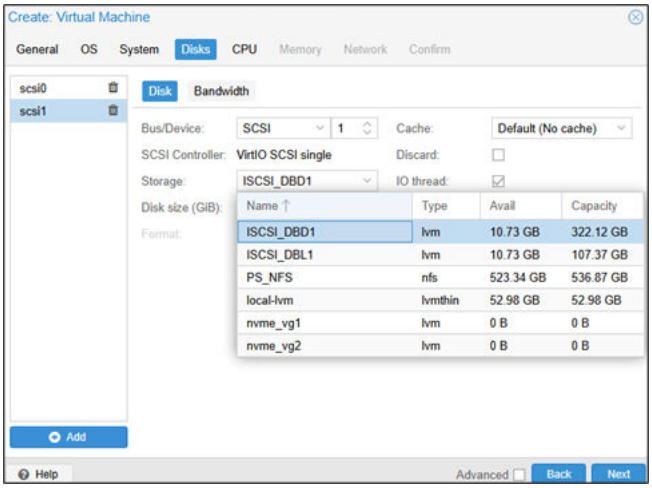

Figure 27. VM disk configuration

5. In the **CPU** tab:
   a. Configure the:
      i. **Sockets**
      ii. **Cores**
      iii. CPU **Type**
   b. Click **Next**.
6. In the **Memory** tab:
   a. Configure the **amount of Memory** to be assigned to the VM.
   b. Click **Next**.
7. In the **Network** tab:
   a. Choose the VM management interface.
   b. Click **Next**.
8. In the **Confirm** tab:
   a. Review the configuration of the new VM.
   b. Click **Finish** to create the VM.

   The newly created VM appears in the resource tree on the left side of the screen.

9. Click the **VM** to see its specifications and options.

   For more information about creating a VM in Proxmox, see Virtual Machines Settings.

VM templates

A template is a fully pre-configured operating system image that can be used to deploy virtual machines.

About this task

Templates are created by converting a VM to a template.

Follow the steps below to create a template:

Steps

1. Install VM with all required drivers and software packages.
2. Remove all user data, passwords, keys, ISO's etc.
3. Power-off the VM.
4. Right-click the VM and select **Convert to template**.
5. Click **Yes**, to confirm the conversion of VM to template.

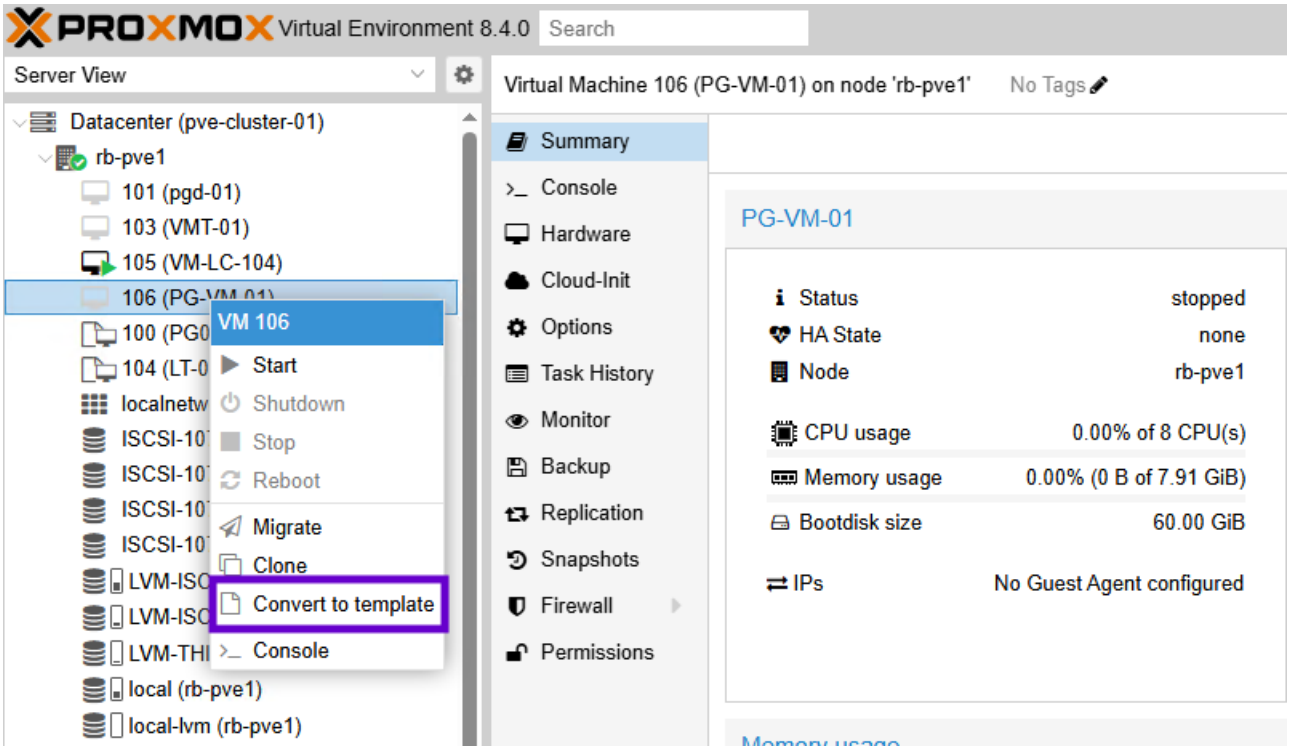

Figure 28. Create VM template

Once the VM is converted, it cannot be started anymore and its icon changes to that of a VM template.

VM cloning

About this task

Proxmox supports two modes of cloning:

Full Clone	A full clone VM is a complete copy and is fully independent from the original VM or VM Template, but it requires the same disk space as the original.
Linked Clone	A linked clone VM requires less disk space but cannot run without access to the base VM Template. Linked Clones works for these storages: files in raw, qcow2, vmdk format (either on local storage or nfs); LVM-thin, ZFS, and so forth . It is not supported with LVM and iSCSI shared storage.

Follow the steps below to create a VM clone from a VM template:

Steps

1. Right-click the VM template and click **Clone**.
2. In the **Clone VM Template** window, complete the following:
   a. Select the **Target node** from the drop-down menu.
   b. Select the **Mode**.
   c. Provide a **Name** for the new VM.
   d. Click **Clone**.

   For more details on templates and clones, see VM Templates and Clones.

VM migration

In a Proxmox cluster, VMs can be migrated from one host to another. Proxmox supports two migration mechanisms:

- Online Migration (Live Migration)
- Offline Migration

Online migration

Online migration is performed when the VM is running and there are no locally bound resources configured (such as devices that are passed through) on it.

Prerequisites

- The VM should not have local resources like any pass-through devices configured on it.
- The hosts are in the same Proxmox VE cluster and the hosts must be configured with CPUs from the same vendor with similar capabilities.
- The hosts have a working and reliable network connection between them.
- Shared storage is accessible by all cluster nodes.

About this task

Online migration allows you to migrate a running VM from one host to another host.

Follow the steps below for VM online migration:

Steps

1. Right-click the running VM and click **Migrate**.
2. In the **Migrate VM** window, select the **Target node** from the drop-down menu and click **Migrate**.

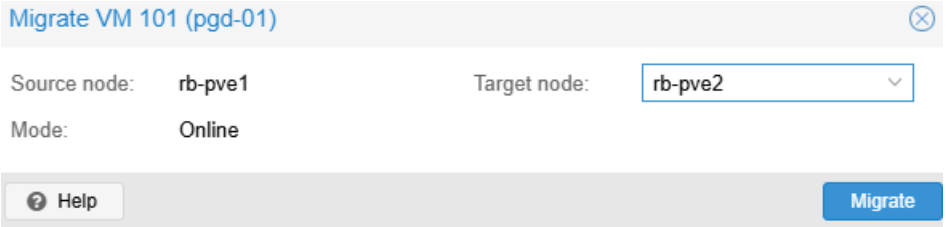

Figure 29. VM Online Migration

The VM migrates to the target node.

Offline Migration

Offline Migration is performed when the VM is in powered off state.

Follow the same procedure as mentioned in the Online migration to complete the offline migration to another node in the cluster.

Storage migration

About this task

Storage migration allows to move a VMs virtual disk to another storage or to another virtual disk format on the same storage.

Follow the steps below for migrating the storage:

Steps

1. Go to the **Virtual Machine** > **Hardware** click the **Hard Disk** to be migrated.
2. From the drop-down menu of **Disk Action**, select **Move Storage**.

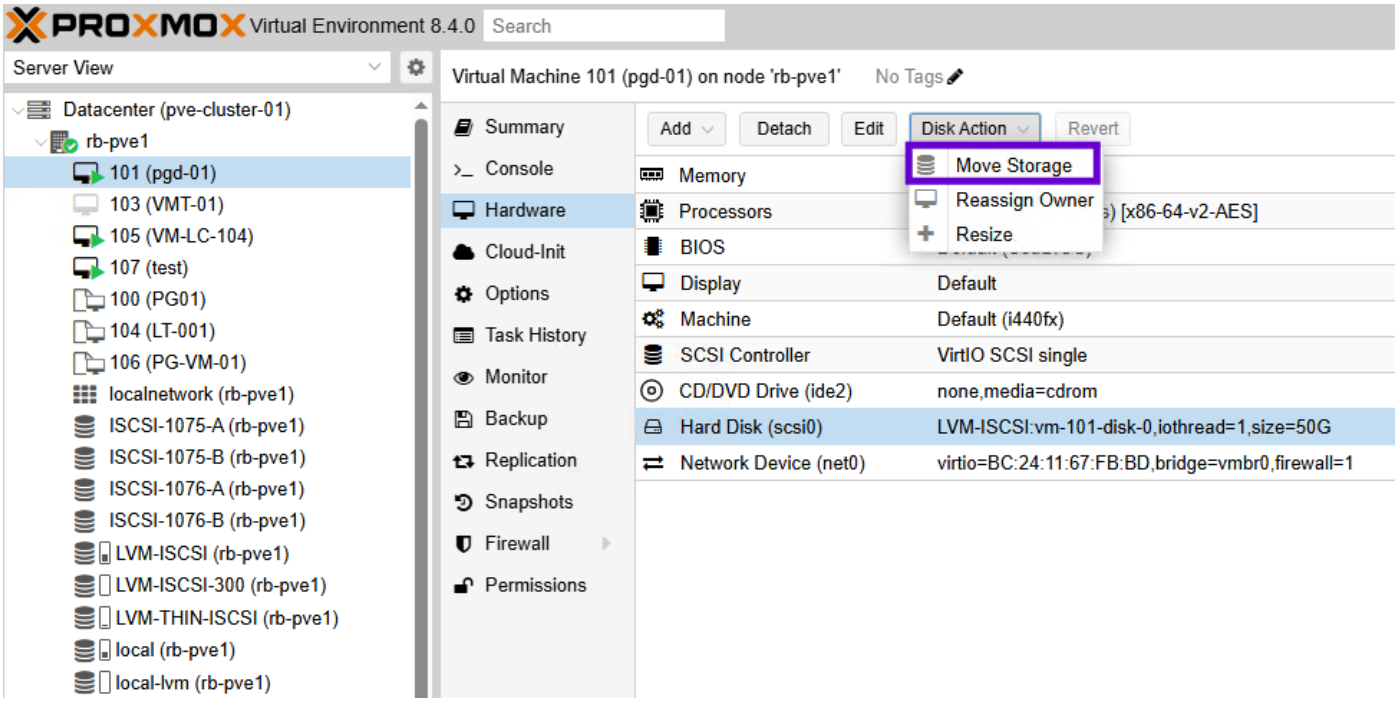

Figure 30. Storage migration

3. In the **Move disk** window, select the **Target Storage**.

   By default, the source disk will be removed and will be displayed as "unused disk" for safety in the VMs **Hardware** list. If you do not need this, just enable "**Delete source**" and click **Move disk**.

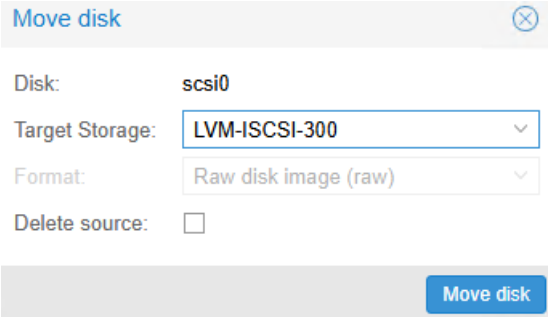

Figure 31. Select target storage for storage migration

4. A **Task viewer** window shows the progress of the operation.

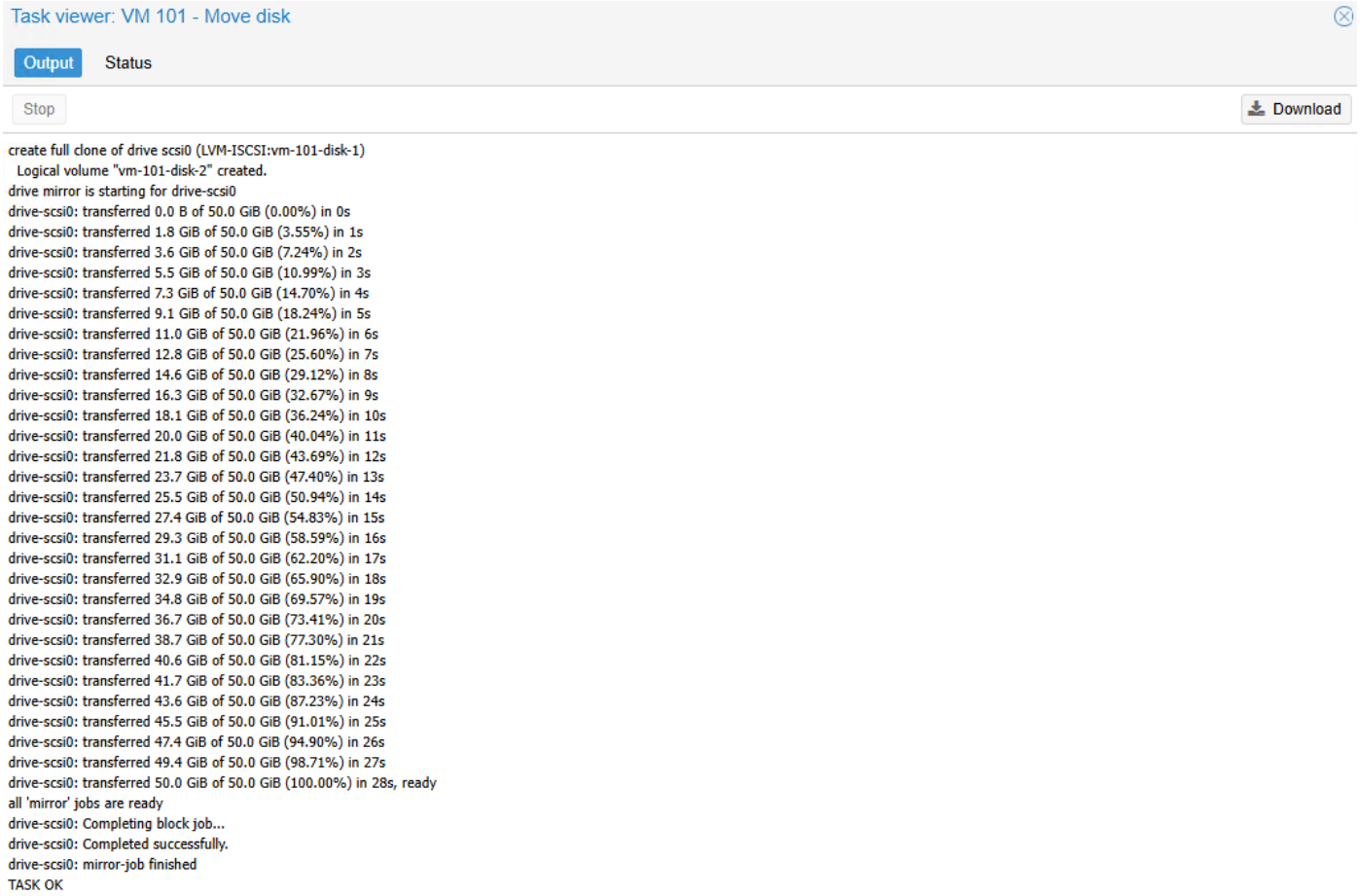

Figure 32. Task viewer window showing the progress of the operation

Once the operation is completed, the virtual disk is moved to the target storage.

VM Snapshots

About this task

A VM snapshot helps to preserve the virtual machine state by including the contents of the virtual machine memory (optional), virtual machine settings, and the state of all the virtual disks. The rollback to a snapshot restores the memory, virtual disks, and all settings of the virtual machine to the state they were in when the snapshot was taken.

NOTE: VM Snapshot is supported only with LVM-thin backend storage.

Follow the steps below to take a snapshot of a VM.

Steps

1. Go to **Datacenter** > **Host** and select the **VM**.
2. Click **Snapshots** and then click **Take Snapshot**.

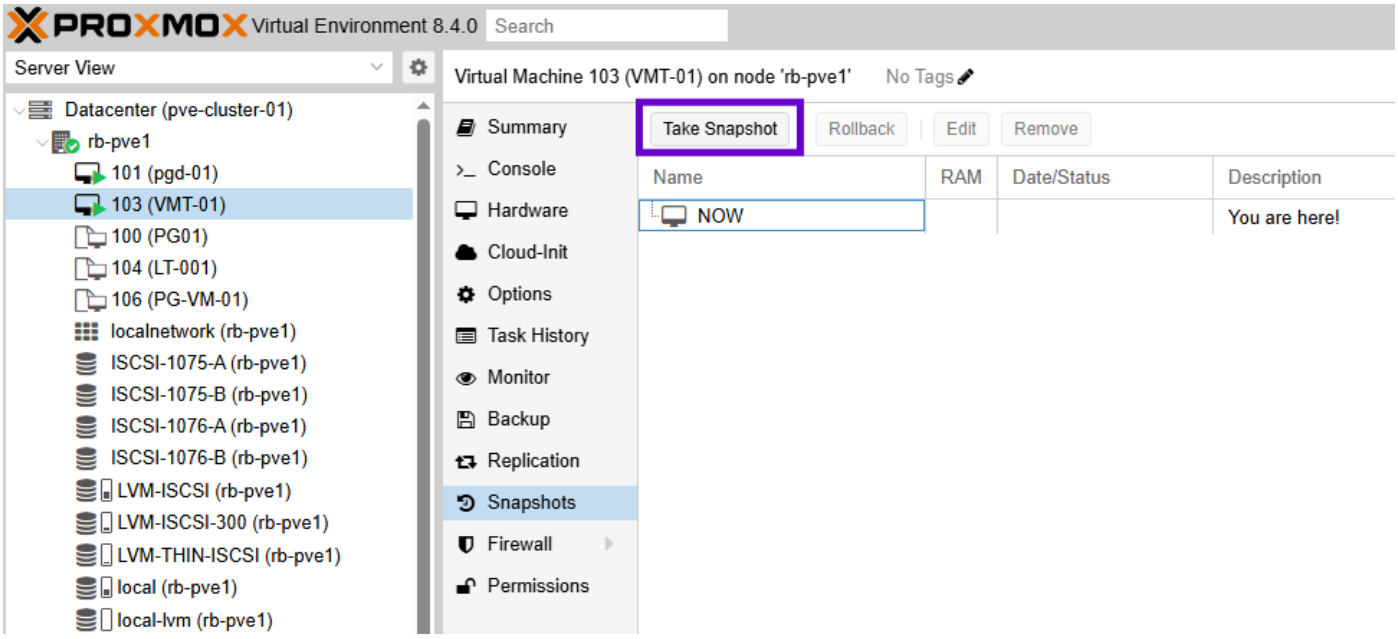

Figure 33. VM Snapshot

3. In the **Create Snapshot** window, provide a **Name** of the snapshot, optionally provide a **Description**, enable **Include RAM** if required and click **Take Snapshot**.

   NOTE: If memory contents are included, the Snapshot operation takes longer to complete and will also require more space.

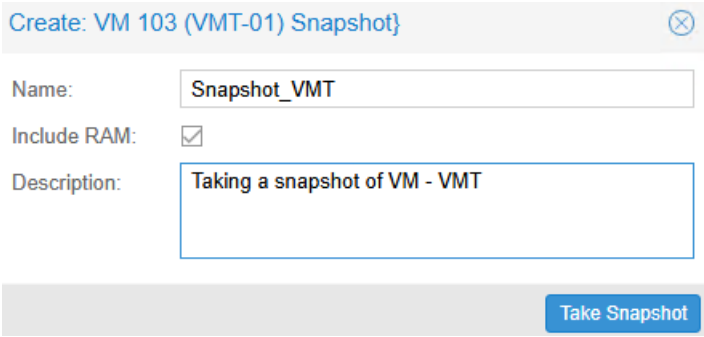

Figure 34. VM Snapshot details

4. The **Task viewer** window shows the progress of the operation.
5. Once the operation is completed, the snapshot is listed in the **Snapshots** window of the VM.

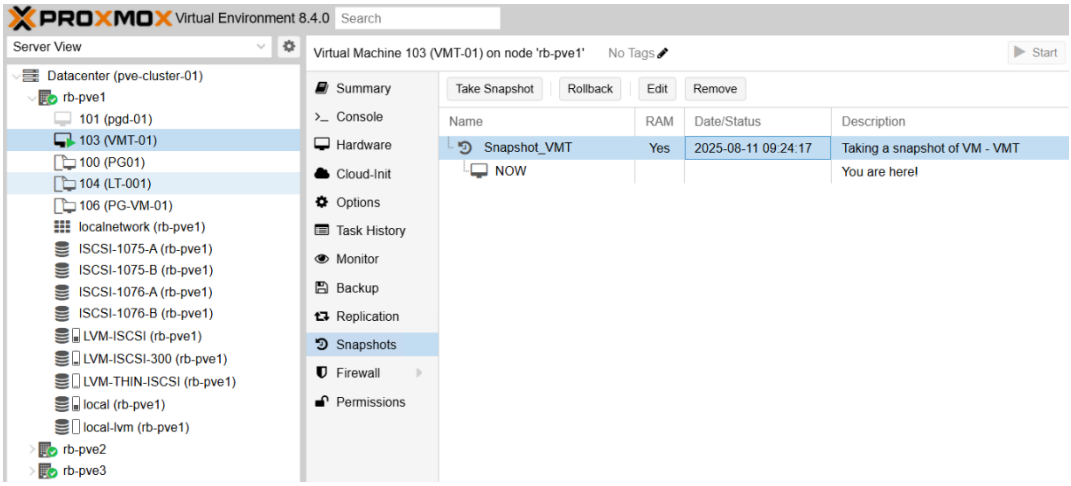

Figure 35. List of Snapshots created

6. To roll back to a snapshot of the VM, select the required snapshot from the list and click **Rollback**.

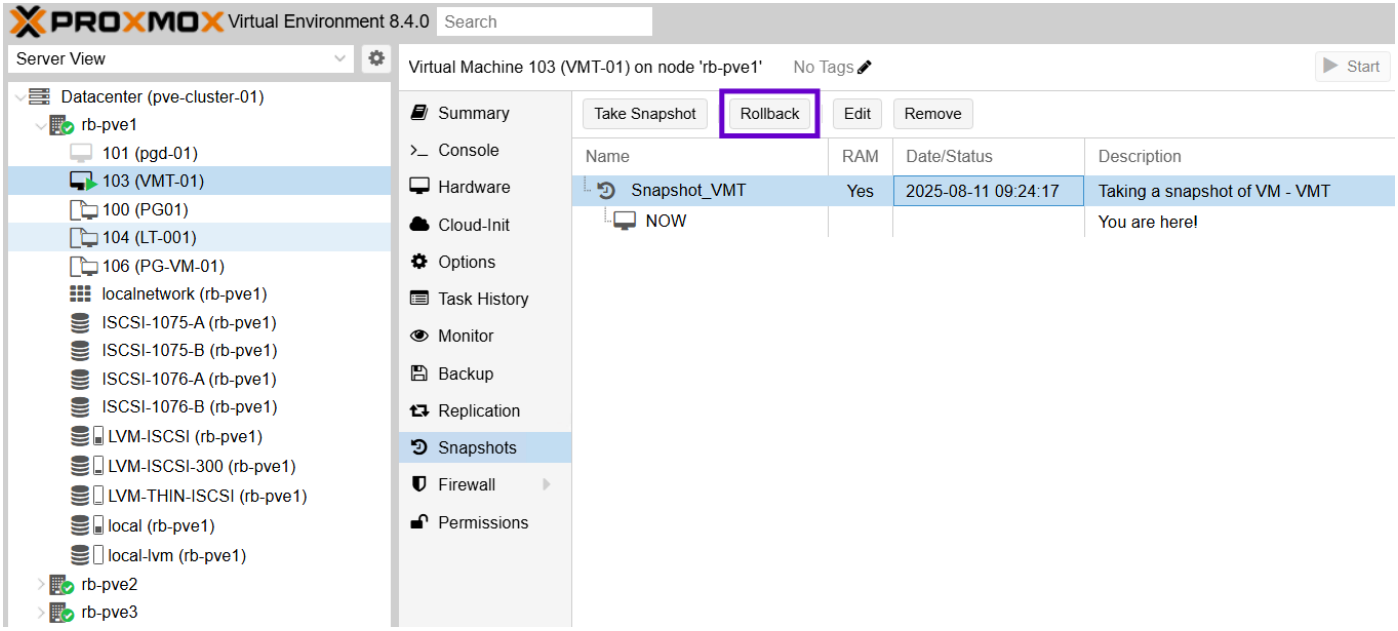

Figure 36. Rollback to a snapshot

7. Click **Yes** to confirm the rollback operation when prompted.

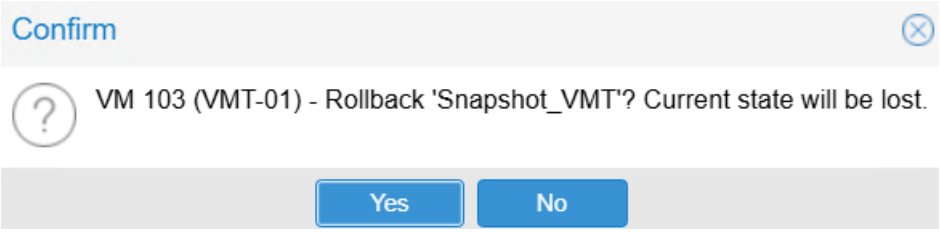

Figure 37. Rollback confirmation

Once the rollback operation is completed, the VM would be in the same state as the snapshot image.

For more details on snapshots in Proxmox, see Live Snapshots.

Deploy PostgreSQL on a virtual machine in Proxmox VE

This section provides instructions to deploy a PostgreSQL database on a VM in Proxmox VE.

Prerequisites

The following requirement must be met before deploying PostgreSQL on the VM:

- In this solution, a Virtual machine (VM) was created with the Red Hat Enterprise Linux 9.2 operating system. The VM is mapped with two LVM disks (iSCSI multipath disks) provisioned by PowerStore system for PostgreSQL database configurations.

For more details on VM resources, see Configuration details.

About this task

Follow the instructions below to deploy PostgreSQL.

Steps

1. Download and install the PostgreSQL software on the virtual machine:

```
# sudo dnf install -y https://download.postgresql.org/pub/repos/yum/reporpms/EL-9-
x86_64/pgdg-redhat-repo-latest.noarch.rpm
# sudo dnf -qy module disable postgresql
# sudo dnf install -y postgresql16-server
```

2. Set up directories and ownership before initializing the PostgreSQL database:

```
# sudo mkfs.xfs /dev/sda
# sudo mkfs.xfs /dev/sdc
# sudo mkdir /mnt/data
# sudo mkdir /mnt/logs
# sudo mount /dev/sda /mnt/data/
# sudo mount /dev/sdc /mnt/logs/
# sudo chown postgres:postgres /mnt/data/
# sudo chown postgres:postgres /mnt/logs/
# sudo chmod 0700 /mnt/data/
# sudo chmod 0700 /mnt/logs
# su - postgres
```

3. Initialize PostgreSQL:

```
$ /usr/pgsql-16/bin/initdb -D /mnt/data/
```

4. Start PostgreSQL:

```
$ /usr/pgsql-16/bin/pg_ctl -D /mnt/data/ -l /mnt/logs/db1.logfile -o "-p 5432" start
```

   The PostgreSQL database server software processes are successfully initialized, running, and listening for the client connectivity at the configured port.

5. Create the user databases with your preferred database name, and verify that the database is created:

```
# sudo -u postgres createdb -h localhost -p 5432 mydb1
# su - postgres
$ psql -p 5432 -d mydb1 -U postgres
psql (16.6)
Type "help" for help.
mydb1=#
```

Results

A new PostgreSQL database mydb1 is created successfully, and the postgres user can log in to it.

Best practices

PowerStore best practices

The following are two best practices that Dell Technologies recommends while using PowerStore:

- PowerStore models support a unified (block and file) or block-optimized (block only) configuration. For a multi-appliance PowerStore T model cluster, only the first appliance in the cluster (the primary appliance) can be configured to support unified storage. Extra appliances added to a cluster are configured automatically as block-optimized during the initial configuration. If file services (NAS) are or may be needed later, select `Unified` during initial configuration. With unified storage configurations, some PowerStore compute and storage resources on the appliances are reserved for file services.
- While using PowerStore iSCSI on the host servers, Dell Technologies recommends that you modify the configuration file (`/etc/iscsi/iscsid.conf`) with the following parameters:

```
node.session.timeo.replacement_timeout = 15
node.startup = automatic
node.session.iscsi.InitialR2T = Yes
node.session.iscsi.ImmediateData = No
```

For more information about PowerStore best practices, see Dell PowerStore: Best Practices Guide.

PostgreSQL configuration

Each PostgreSQL database has its own configuration files. Dell Technologies recommends that you carefully review and edit the tunable parameters of the `postgresql.conf` configuration file to meet the specific needs of the database workload.

Values for configurable parameters like `data_directory`, `log_directory`, `port`, `max_connections`, `shared_buffers`, `min_wal_size`, and `max_wal_size` can be specified based on the system configuration. The recommended value for these tunable parameters depends on the overall system resources available.

Online tools can be used for tuning the PostgreSQL parameters. For more information, see PostgreSQL Tunable parameters.

Conclusion

This white paper describes the deployment of Proxmox VE. It provides instructions to configure iSCSI, NVMe/TCP, and NFS on Proxmox VE nodes to access PowerStore block and file resources. The paper also describes various virtualization tasks on virtual machines in Proxmox VE. It also documents the deployment of a PostgreSQL database using a PowerStore system on a VM deployed in Proxmox VE.

PostgreSQL is an advanced database system that can support different types and sizes of workloads with a high performance and a robust set of features.

Proxmox VE offers a balance of features, ease of use, and cost-effectiveness, making it ideal for deploying PostgreSQL database or other enterprise workloads.

References

Dell Technologies documentation

The following Dell Technologies documentation provides additional information. Access to these documents depends on your login credentials. If you do not have access to a document, contact your Dell Technologies representative.

- Dell PowerStore: Info Hub - Product Documentation and Videos
- Dell Technologies Info Hub for PowerStore
- Dell PowerStore: Introduction to the Platform
- Dell PowerStore Manager Overview
- Dell PowerStore: Best Practices Guide

PostgreSQL documentation

The following PostgreSQL documentation provides additional information.

- PostgreSQL documentation
- Installing PostgreSQL

Proxmox documentation

The following Proxmox documentation provides additional information.

- Proxmox documentation

Appendix A

Configuration details

The following tables show the hardware and software configuration of the storage and host servers that are used in this solution.

Table 5. PowerStore hardware and software details

Hardware	Configuration
PowerStore model	Dell PowerStore 5200T
CPU	4 x Intel® Xeon® CPUs 96 cores, 2.2 GHz
Memory (GiB)	1152 GiB
NIC	4-port 25 GbE I/O Module
Storage	• 4 x 8.5 GB NVMe NVRAM • 12 x 3.84 TB NVMe SSD TLC
Dell PowerStore Manager	4.0

Table 6. Proxmox host server hardware and software details

Hardware	Configuration
Server model	Dell PowerEdge server
CPU	2 x Intel® Xeon® CPUs @ 3.00 GHz
Memory (GiB)	512 GiB
NIC	4 x 10 GbE
Storage	BOSS 2 x M.2 SSDs 240 GB
Operating system	Proxmox VE-8.3.0/Debian GNU/Linux 12

Table 7. Proxmox VE virtual machine hardware and software details

Hardware	Configuration
CPU	4 x vCPU
Memory (GB)	16 GB RAM
NIC	1 x 10 GbE
Storage	• Operating system disk: 30 GB • Data disk1: 300 GB • Data disk2: 100 GB
Operating system	Red Hat Enterprise Linux 9.2

Proxmox network configuration file

An example Proxmox network configuration file (`/etc/network/interfaces`) is provided below.

```
# cat /etc/network/interfaces
auto lo
iface lo inet loopback

auto p1p1
iface p1p1 inet manual
```

```
iface p1p2 inet manual
       mtu 9000

auto p2p1
iface p2p1 inet manual

iface p2p2 inet manual
       mtu 9000

auto bond0
iface bond0 inet manual
       bond-slaves p1p1 p2p1
       bond-miimon 100
       bond-mode active-backup
       bond-primary p1p1
       mtu 9000

auto vmbr0
iface vmbr0 inet manual
       bridge-ports bond0
       bridge-stp off
       bridge-fd 0
       bridge-vlan-aware yes
       bridge-vids 2-4094
       mtu 9000

auto vmbr0.132
iface vmbr0.132 inet static
       address 192.168.132.199/24
       gateway 192.168.132.254
       mtu 9000

auto vmbr0.133
iface vmbr0.133 inet static
       address 192.168.133.199/24
       gateway 192.168.133.254
       mtu 9000

auto bond1
iface bond1 inet manual
       bond-slaves p1p2 p2p2
       bond-miimon 100
       bond-mode active-backup
       bond-primary p1p2
       mtu 9000

auto vmbr1
iface vmbr1 inet manual
       bridge-ports bond1
       bridge-stp off
       bridge-fd 0
       bridge-vlan-aware yes
       bridge-vids 2-4094
       mtu 9000

auto vmbr1.1075
iface vmbr1.1075 inet static
       address 192.168.175.199/24
       gateway 192.168.175.254
       mtu 9000

auto vmbr1.1076
iface vmbr1.1076 inet static
       address 192.168.176.199/24
       gateway 192.168.176.254
       mtu 9000

source /etc/network/interfaces.d/*
```